

Volumetric-Aware Grasp Obstruction Reasoning for Manipulation in Clutter

Anonymous Author(s)

Affiliation

Address

email

Abstract: In cluttered environments, grasping a target object often requires re-locating surrounding obstacles, but identifying which objects to remove is challenging: the obstacles depend on the chosen $SE(3)$ grasp pose and the robot’s full embodiment, and many regions are occluded under partial observation. We address this with grasp obstruction reasoning—predicting which objects collide with the robot at a target grasp—and propose Volumetric Obstruction Reasoner for Manipulators (VORM). VORM performs object-wise obstruction prediction by jointly reasoning over point clouds of the segmented scene and the manipulator’s full arm-gripper volume, with cross-attention modeling their local geometric relation at the target grasp. To train VORM, we build a procedural pipeline on top of FetchBench that generates cluttered scenes at three clutter levels and annotates per-grasp obstruction labels. VORM outperforms baselines on obstruction and feasibility reasoning in highly cluttered, partially observed scenes and improves success rates on target-retrieval tasks.

Keywords: Obstruction reasoning, Point cloud learning, Grasping in clutter

1 Introduction

Grasping a target object in cluttered environments has been widely studied in robotic manipulation. When the target is obstructed by surrounding objects, some of them should be relocated first to enable a collision-free grasp—yet identifying an efficient removal strategy is inherently difficult. This is because the obstacles to be removed are tightly coupled with how the target is grasped, and depend on the robot’s embodiment and configuration. This coupling is particularly complex for everyday objects, whose diverse, multi-modal grasp distributions over $SE(3)$ yield qualitatively different removal strategies across grasp modes. The challenge is further compounded by partial observability, where analytic collision checkers cannot reliably identify collisions in occluded regions [1].

A common approach for grasping a target in clutter is to reason about *grasp feasibility*—predicting whether a given grasp can be executed without collision. However, when every candidate grasp is in collision, feasibility alone gives no guidance on which objects to remove. This motivates reasoning about *grasp obstruction*—which objects block the target grasp, which has proven effective for Task and Motion Planning (TAMP) problems such as target retrieval [2, 3, 4, 5]. Specifically, we define obstruction reasoning as determining which surrounding objects collide with the robot at a grasp pose, assuming that removing them enables a collision-free grasp. Since a planner removes the predicted objects, accuracy is critical: errors lead to unnecessary or repeated removals.

To make obstruction reasoning practical across the diverse objects, scenes, and partial observations of real-world clutter, we identify several requirements a reasoner should satisfy. First, it should reason over the robot’s full 3D volume, including the arm and not just the gripper, for accurate collision prediction. Second, it should operate from partial observations without object meshes, since occlusion and incomplete geometry are common in clutter. Third, it should predict collisions for

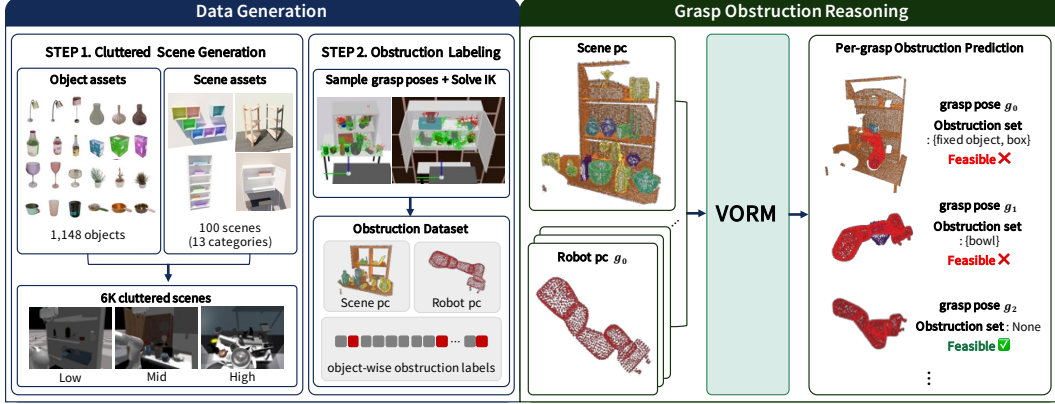


Figure 1: System overview. **(Left)** We procedurally generate cluttered scenes and annotate obstruction labels. **(Right)** Given a segmented scene point cloud and a robot point cloud induced by a grasp g , VORM predicts the obstruction set; different grasps yield different obstructions and feasibility.

arbitrary SE(3) grasps, since everyday objects and surrounding fixed structures such as shelves, drawers, and desks induce diverse, multi-modal grasp distributions. Finally, it should generalize across varying clutter levels and robot embodiments. To the best of our knowledge, no existing approach satisfies all of these requirements: prior methods either assume full observability [3, 6], restrict grasps to fixed directions [3, 4, 5], reason about occlusion only in 2D [7, 8, 9], or do not easily transfer across embodiments [10, 11].

We propose Volumetric Obstruction Reasoner for Manipulators (VORM), a model that predicts which objects obstruct a target grasp in a cluttered scene. The core design of VORM is to reason about collisions directly between point clouds of the robot and the scene. It represents the robot as the space occupied by the arm and gripper at the target grasp, and the scene as per-object segments. In a shared space, it predicts object-wise collisions via attention over the point-to-point relationships between the robot and each object. This design lets VORM meet all the requirements above. Representing the full volume of the robot as a point cloud makes prediction accurate and applicable to any SE(3) grasp or embodiment. As VORM reasons over the observed scene points, it handles partial observations and generalizes across diverse cluttered scenes. We further build a cluttered scene dataset on FetchBench [12] to train VORM, leveraging its diverse objects and scenes while extending them with denser arrangements at varying clutter levels and adding per-grasp obstruction labels.

Our contributions are as follows:

- We propose a grasp obstruction reasoner that, given an arbitrary SE(3) grasp in a cluttered scene, predicts which objects to remove by jointly reasoning over point clouds of the scene and the robot—represented by the full arm-gripper volume.
- We introduce VORM, a model that enables accurate obstruction prediction through point-wise cross-attention between the robot and segmented scene point clouds.
- Building on diverse object and scene assets, we propose a procedural pipeline that generates cluttered scenes at varying clutter levels with object-wise obstruction labels.
- We demonstrate that VORM outperforms baselines on obstruction and feasibility reasoning in highly cluttered, unstructured environments under a single camera setting. Moreover, we show that accurate obstruction reasoning leads to higher success rates in target retrieval.

2 Related Works

Grasp Feasibility and Obstruction Reasoning. Feasibility reasoning in cluttered scenes has been widely studied to accelerate TAMP, predicting whether a target can be grasped from scene representations such as depth images [13, 14, 15, 16] or voxel grids [17]. These methods, however, consider

only discrete grasps on simplified geometries, rather than the arbitrary SE(3) grasps everyday objects require. Beyond feasibility, several works reason about which objects obstruct the target grasp. Some approximate obstruction using the gripper footprint [4, 18, 5], while image-based methods infer obstruction from amodal occlusion segmentation [19, 20, 7] or the spatial reasoning of vision-language models [21, 22, 8, 9]. Nonetheless, these 2D-based methods cannot accurately capture the 3D collision that determines obstruction. Geometric Reasoning Network (GRN) [3] instead grounds obstruction in 3D object geometry, jointly predicting feasibility and obstruction, but assumes full observability and predefined grasp modes, limiting real-world applicability.

Collision Handling in Clutter. Identifying obstructing objects is closely tied to handling collisions in clutter. Analytic collision checkers such as Flexible Collision Library (FCL) [6] exactly identify whether and which objects obstruct a grasp, given complete object geometry. In practice, however, only partial observations are available, motivating learning-based methods that operate on partial point clouds. One line generates 6-DoF grasps in clutter by jointly considering grasp quality and collision-freeness, where collision is determined by the gripper’s geometry against the observed scene point cloud [23, 24, 25, 10, 11]. Another line trains neural collision predictors that estimate collision between a moving object and the scene, both as point clouds [2, 26, 1], generalizing across objects and scenes; trained on ground-truth collision labels, they infer collisions even in unobserved regions. By reasoning in 3D, these methods are accurate and effective under partial observation, yet they neglect the robot’s occupied space and obstruction reasoning. We therefore represent both the scene and the entire robot, not just the gripper, as point clouds, and reason about their collisions to predict object-wise obstructions. This design inherits the partial observation robustness of prior predictors and generalizes across embodiments.

Cluttered Scene Benchmarks. Existing benchmarks for cluttered scenes often trade diversity against clutter density. FetchBench [12], for instance, spans 5.5k objects across 13 procedural scenes (desks, shelves, drawers, etc.) but places only a few objects per scene, while other benchmarks [27, 28] achieve denser arrangements within a narrower range of objects or scenes. Some real-world datasets, such as GraspClutter6D [29], attain both but are not procedurally extensible. Measures of clutter also vary across works, ranging from visual occlusion ratio [28, 30, 31] to the number of objects [27, 3, 32] to feature congestion [33]. To bridge this gap, we propose a procedural pipeline that retains FetchBench’s object and scene diversity while generating denser arrangements at varying clutter levels, measured by occupancy ratio.

3 Data Generation

We generate training data for manipulation in diverse cluttered scenes in simulation. Leveraging diverse object and scene assets from FetchBench [12], we propose a procedural pipeline that arranges objects into cluttered scenes at three levels—low, medium, and high—to ensure data diversity and enable evaluation across difficulty levels. In addition, for each cluttered scene, we sample grasp candidates and annotate each with an object-wise obstruction label for training.

Object and scene assets. We build on assets from FetchBench [12], which integrate ACRONYM [34] and Infinigen [35] to provide 5,544 objects and 13 categories of procedural scenes. Each object is annotated with stable poses and grasp poses following ACRONYM, and each scene with one or more support surfaces for object placement. From these objects, we filter out (i) categories semantically unsuitable for placement on a support surface (e.g., furniture, architectural elements), (ii) objects that fail to remain static under physics simulation, and (iii) objects with insufficient collision-free grasps, under a gripper-only collision check. This yields a final set of 1,148 objects. For scenes, we use 100 procedurally generated base scenes spanning all 13 categories.

Cluttered Scene Generation. Following FetchBench [12], we generate cluttered scenes (Fig. 2) by sequentially placing sampled objects at stable poses on a base scene’s support surfaces. We extend this with controllable clutter at three levels, parameterized by the *occupancy rate* $\rho =$

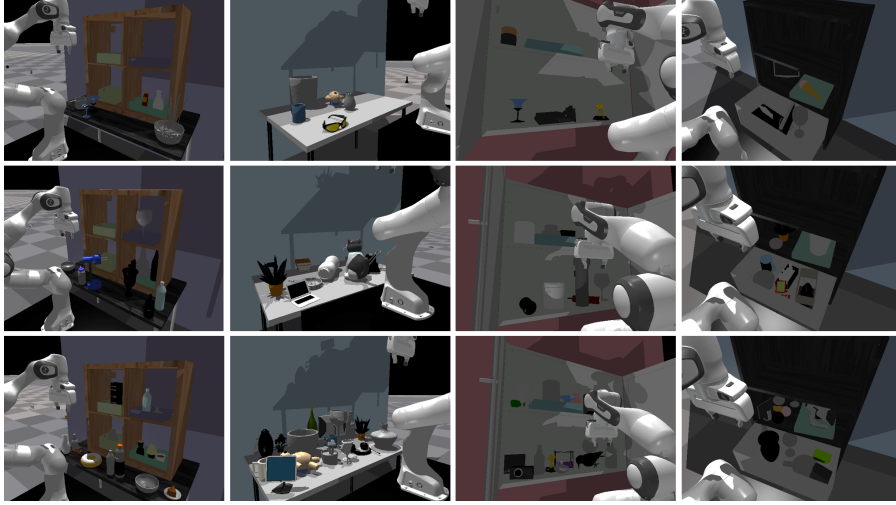


Figure 2: Generated scenes across three clutter levels and four scene categories. Top to bottom: low, medium, high clutter. Left to right: CellShelfDesk, Desk, DoubleDoorCabinet, DrawerShelf.

118 $\sum_{i=1}^K A_i^{\text{obj}} / A^{\text{scene}}$, where K is the number of objects, A_i^{obj} is object i 's footprint (base area of its
 119 yaw-aligned bounding box), and A^{scene} is the total support-surface area. Because the same ρ can
 120 arise from few large or many small objects, we additionally restrict per-scene object scale rela-
 121 tive to A^{scene} . For each level, we constrain ρ to $[15, 20]$ (low), $[30, 35]$ (medium), or $[45, 50]$ (high)
 122 via rejection sampling. We instantiate scenes in Isaac Gym [36] with a 7-DOF Franka Panda arm
 123 and gripper, with a single side-mounted camera providing partial point-cloud observations. Robot
 124 and camera poses are randomized within suitable ranges. We split the 100 base scenes into 80/20
 125 train/test sets, generating 60 cluttered scenes per base scene (20 per level) for 6,000 total.

126 **Obstruction Labeling** For each cluttered scene, we collect 100–200 grasp configurations, each
 127 comprising a robot point cloud, a scene point cloud, and an object-wise obstruction label, yielding
 128 687K total. We sample grasp poses from the annotated grasps of all K objects and solve inverse
 129 kinematics via cuRobo [37] for one configuration per grasp, discarding unreachable poses. At each
 130 configuration, FCL [6] checks collisions against $K+1$ objects—the K movable objects individually
 131 plus the rest as one fixed object—defining the binary label. The scene point cloud (from the single
 132 camera) follows the same $K+1$ partition. The robot point cloud is sampled from the robot mesh.

133 4 Volumetric Obstruction Reasoner for Manipulators (VORM)

134 VORM predicts which objects obstruct a candidate grasp in a cluttered scene. The key idea is to
 135 represent both the observed scene and the manipulator volume induced by the grasp pose as point
 136 clouds, and to learn their geometric interaction for object-wise obstruction prediction.

137 4.1 Problem Formulation

138 We formulate obstruction reasoning as estimating, for each object, the probability that it collides
 139 with the robot at a grasp pose $g \in SE(3)$, conditioned on a segmented scene point cloud and a robot
 140 point cloud induced by g . Section 3 describes how both point clouds and the obstruction labels are
 141 produced.

142 We segment the scene into K movable objects with IDs $0, \dots, K-1$, and treat all remaining points
 143 as a single fixed object with ID K . This yields a segmented point cloud of N_s points, each with a
 144 coordinate $\mathbf{x}_i^s \in \mathbb{R}^3$ and an object ID $a_i^s \in \{0, \dots, K\}$, forming

$$\mathcal{S} = \{(\mathbf{x}_i^s, a_i^s)\}_{i=1}^{N_s}. \quad (1)$$

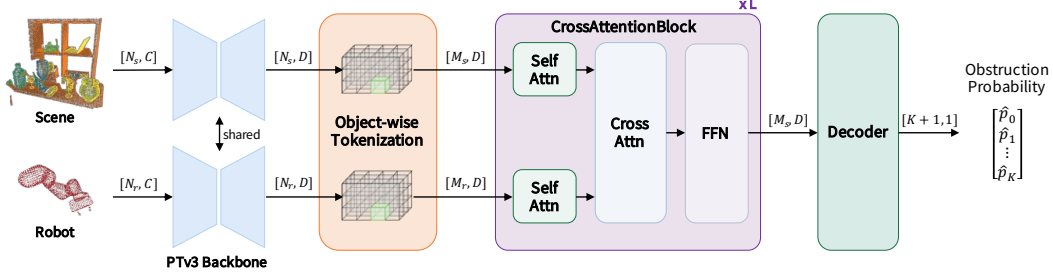


Figure 3: Overall architecture of VORM. A PTv3 backbone and object-wise tokenization turn the scene and robot point clouds into object-wise tokens. L cross-attention blocks fuse the scene and robot tokens, and a decoder predicts a per-object obstruction probability for each scene object.

For the grasp g , we obtain a joint configuration q_g , assuming a single IK solution per grasp. We then represent the robot at q_g , including both the arm and the gripper, as a point cloud of N_r points, each with a coordinate $\mathbf{x}_j^r \in \mathbb{R}^3$ and a dedicated object ID $a_j^r = K+1$, giving

$$\mathcal{R}_g = \{(\mathbf{x}_j^r, a_j^r)\}_{j=1}^{N_r}. \quad (2)$$

The binary obstruction label $y_k \in \{0, 1\}$ indicates whether the robot at q_g collides with object k . Formally, the goal is to predict from the two point clouds, for the grasp g , the obstruction probability of each of the $K+1$ objects,

$$\hat{p}_k = P(y_k = 1 \mid \mathcal{S}, \mathcal{R}_g), \quad k = 0, \dots, K. \quad (3)$$

4.2 Network Architecture

VORM processes the scene and robot point clouds through point encoding, object-wise tokenization, and cross-attention to predict per-object obstruction (Fig. 3).

Point Feature Encoding VORM encodes the scene and robot point clouds into point-wise features through a weight-shared Point Transformer V3 (PTv3) [38] backbone, applied separately to each cloud. Before encoding, a learnable embedding φ maps the object ID of each point in $\mathcal{S}$ and $\mathcal{R}_g$ to a feature vector $f = \varphi(a)$. From these $(\mathbf{x}, f)$ pairs, the backbone produces point-wise features $\{h_i^s\}_{i=1}^{N_s}$ for the scene and $\{h_j^r\}_{j=1}^{N_r}$ for the robot, where $h_i^s, h_j^r \in \mathbb{R}^D$. Unlike point-grouping or sparse-convolution backbones [39, 40] that aggregate features within fixed local neighborhoods, PTv3 integrates context over a wider receptive field, so each point feature reflects its whole object’s geometry rather than a local patch. This is especially valuable for the partially observed scene cloud, where broader context helps infer occluded geometry.

Object-wise Tokenization To produce object-wise obstruction predictions that depend on local geometric interaction, VORM needs tokens that preserve both local geometric structure and object identity, while staying compact for tractable attention. We introduce an object-wise voxel pooling tokenizer that converts the point-wise features $\{h_i^s\}_{i=1}^{N_s}$ and $\{h_j^r\}_{j=1}^{N_r}$ into tokens, each summarizing a voxel-local region of a single object. We voxelize space and group points by their (voxel index, object ID) pair; each group thus has a single object ID by construction. For each group, MaxPool over features gives the token feature, AvgPool over coordinates gives the token coordinate, and a sinusoidal positional encoding is applied to it. This yields scene tokens $\{v_m^s\}_{m=1}^{M_s}$ and robot tokens $\{v_n^r\}_{n=1}^{M_r}$, with $v_m^s, v_n^r \in \mathbb{R}^D$. Each scene token v_m^s inherits the object ID $a_m^s \in \{0, \dots, K\}$ of its points, for later per-object aggregation.

Cross-Attention for Prediction Predicting obstruction for each scene object requires reasoning about its geometric interaction with the robot at q_g . We model this cross-cloud interaction with DETR-style [41] cross-attention. Scene tokens $\{v_m^s\}_{m=1}^{M_s}$ serve as queries and robot tokens $\{v_n^r\}_{n=1}^{M_r}$ as keys and values, so each scene token gathers robot-side collision evidence for its region. The

model stacks L such blocks, each applying scene self-attention, robot self-attention, cross-attention, and feed-forward layers, with sinusoidal positional encodings added to queries and keys but not values. After L blocks, a 4-layer MLP maps the updated scene tokens $\{\bar{v}_m^s\}_{m=1}^{M_s}$ to per-token obstruction logits $\{\ell_m^s\}_{m=1}^{M_s}$, with $\bar{v}_m^s \in \mathbb{R}^D$ and $\ell_m^s \in \mathbb{R}$. The logits are then aggregated per object by MaxPool, $\ell_k = \max_{m: a_m^s=k} \ell_m^s$, and a sigmoid yields the object-wise obstruction probability $\hat{p}_k = \sigma(\ell_k)$ of Eq. 3.

4.3 Training Objective

VORM is trained with a weighted sum of a binary cross-entropy (BCE) loss $\mathcal{L}_{\text{BCE}}$ and a pairwise ranking loss [42] $\mathcal{L}_{\text{rank}}$,

$$\mathcal{L} = \lambda_{\text{bce}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}. \quad (4)$$

The BCE loss provides direct per-object supervision by treating each object’s obstruction as an independent binary classification,

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{K+1} \sum_{k=0}^K [y_k \log \hat{p}_k + (1 - y_k) \log(1 - \hat{p}_k)]. \quad (5)$$

BCE supervises each object independently but does not enforce any margin between obstructing and non-obstructing objects within a scene. We therefore add a pairwise ranking loss that encourages obstructing object logits to exceed non-obstructing ones by a margin γ , computed only on scenes containing both. With $\mathcal{P} = \{k : y_k = 1\}$ and $\mathcal{N} = \{k : y_k = 0\}$ denoting these two object groups,

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}||\mathcal{N}|} \sum_{j \in \mathcal{P}} \sum_{k \in \mathcal{N}} \max(0, \gamma - (\ell_j - \ell_k)). \quad (6)$$

5 Experiment

We investigate the following questions in our experiments. **(Q1)** Without explicit object geometry, can VORM predict obstruction and feasibility more accurately than prior baselines? **(Q2)** How does each of VORM’s core components contribute to this accuracy? **(Q3)** Does VORM’s obstruction reasoning improve downstream TAMP target retrieval?

5.1 Quantitative Results

We evaluate two tasks: *obstruction* and *feasibility* reasoning. Obstruction reasoning determines whether the robot collides with each object in the scene, whereas feasibility reasoning determines whether it collides with none. We compare against three baselines. All methods are evaluated on the same test dataset (Section 3) with point cloud inputs, differing only in subsequent processing (e.g., meshing or merging); GRN instead uses an abstracted geometric representation of each object.

- **MC+FCL**: FCL [6] performs collision detection between meshes. We therefore convert each segmented scene point cloud into a mesh via Marching Cubes (MC) and check its collision with the known robot mesh, yielding object-wise obstruction predictions.
- **GRN** [3]: GRN takes a scene graph—nodes are objects with edges encoding inter-object proximity—and uses a GNN to predict obstruction and feasibility with separate modules. It approximates each object as a yaw-aligned bounding box, restricts grasps to five predefined directions, and assumes the full object geometry is known.
- **CabiNet** [1]: CabiNet (CBN) predicts whether a grasped object collides with the scene from their point clouds. We assess grasp feasibility by replacing the object with the robot point cloud against the merged scene point cloud. It predicts feasibility only, not per-object obstruction.

In summary, VORM, MC+FCL, and GRN produce obstruction predictions, while CBN predicts feasibility only. For VORM and MC+FCL, feasibility is max-aggregated from the obstruction predictions; GRN and CBN predict feasibility directly.

Table 1: Obstruction and feasibility performance over all clutter levels (Total) and by clutter level (Low, Mid, High). P and R denote precision and recall. Best per column in **bold**.

Task	Method	Total			Low	Mid	High
		P↑	R↑	F1↑	F1↑	F1↑	F1↑
Obstruction	VORM (Ours)	0.921	0.911	0.923	0.935	0.923	0.912
	MC+FCL	0.721	0.931	0.813	0.827	0.815	0.805
	GRN	0.449	0.241	0.314	0.316	0.321	0.305
Feasibility	VORM (Ours)	0.933	0.912	0.923	0.933	0.922	0.913
	MC+FCL	0.965	0.372	0.537	0.511	0.540	0.557
	GRN	0.310	0.391	0.346	0.327	0.350	0.355
	CabiNet	0.401	0.219	0.283	0.323	0.279	0.251

Table 2: Ablation results. All models are trained on 30% of the training data; numbers are not directly comparable to Table 1. Best per column in **bold**.

Task	Method	Total		
		P↑	R↑	F1↑
Obstruction	VORM (Ours)	0.941	0.858	0.897
	w/o object-wise grouping	0.883	0.835	0.859
	w/o cross-attention	0.809	0.269	0.404
	w/o rank loss	0.926	0.803	0.860

Table 1 reports obstruction and feasibility performance overall (Total) and by clutter level (Low, Mid, High). VORM (Ours) achieves the highest F1 on both tasks across all clutter levels, with a gradual decline as clutter rises that reflects the increasing difficulty. MC+FCL stays moderately behind VORM on obstruction, but falls far short on feasibility. Collision over-prediction explains this gap: Marching Cubes overestimates the volume each object occupies, which shows as low precision in obstruction. GRN and CabiNet both perform poorly, each from a mismatch with our setting: GRN approximates objects as boxes and grasps as a few fixed types, while CabiNet, built for object-scene collision is applied here to out-of-distribution robot queries across more diverse scenes.

5.2 Ablation Study

We analyze the contribution of VORM’s three main components: (i) object-wise tokenization, (ii) the cross-attention, and (iii) the ranking loss. All models are trained on 30% of the training data. Unlike Table 1, we report only obstruction metrics, the direct network output; feasibility is a derived aggregation. We construct three variants of VORM (Ours):

- **w/o object-wise grouping:** the object-wise voxel pooling tokenizer groups points by voxel index alone, instead of by (voxel index, object ID) pairs.
- **w/o cross-attention:** inspired by [1], we replace cross-attention with a late-fusion baseline: robot tokens are pooled into a single global vector and concatenated to the scene token.
- **w/o rank loss:** we train with only the BCE loss $\mathcal{L}_{\text{BCE}}$ in Eq. 4, removing the ranking loss $\mathcal{L}_{\text{rank}}$.

As shown in Table 2, VORM (Ours) outperforms all variants, confirming that each component contributes. The largest drop comes from w/o cross-attention: F1 falls from 0.897 to 0.404, with recall dropping from 0.858 to 0.269. Compressing robot tokens into a single global token removes the per-token robot features that cross-attention can selectively gather, causing the model to miss many collisions and suggesting that token-level scene-robot interactions matter for collision prediction. w/o object-wise grouping reduces F1 from 0.897 to 0.859, as points from different objects sharing a voxel are merged into one token, entangling per-object features. w/o rank loss (F1 0.860) shows that the ranking loss, which enlarges the score gap between obstructing and non-obstructing objects, provides supervision complementary to BCE.

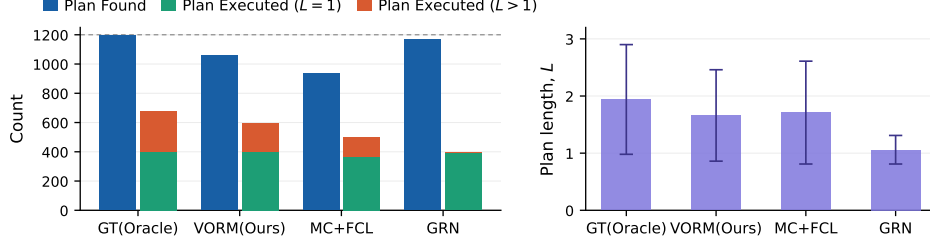


Figure 4: Target retrieval results over $n = 1200$ scenes per method (dashed line). **(Left)** Plan Found: valid plan exists; Plan Executed ($L = 1/L > 1$): execution succeeds at $L = 1$ (direct grasp) or $L > 1$ (after relocation). **(Right)** Plan length L (mean and std), the total picks in the plan.

243 5.3 Simulation Result

244 We examine how obstruction reasoning quality affects downstream target retrieval, a TAMP prob-
 245 lem. Holding the planner fixed, we vary only the obstruction reasoning method and evaluate the
 246 resulting plans and their execution in simulation. The compared methods and test dataset match
 247 those of Table 1. We additionally include GT (Oracle), where ground-truth labels replace model
 248 predictions. Following [5], the target in each scene is the object requiring the most removals before
 249 being graspable; this target is shared across methods. Each method is evaluated through a three-stage
 250 pipeline. (1) *Obstruction reasoning* runs on the initial scene to predict object-wise obstruction for
 251 every candidate grasp. (2) *Relocation planning* uses the planner of [5], which compresses obstruc-
 252 tion predictions into a logical formula and runs backward-chaining search to produce a relocation
 253 plan—a sequence of objects to grasp with their grasp poses. Targets obstructed by the fixed object
 254 (absent in the original setting) are excluded from the formula. (3) *Execution* uses cuRobo [37] to
 255 generate and run trajectories for the relocation plan.

256 Figure 4 reports results over 1,200 scenes; GT (Oracle) serves as an upper-bound reference rather
 257 than a competing baseline, and the outcomes consistently reflect Table 1’s prediction-quality differ-
 258 ences. VORM, with the highest F1, achieves the highest execution success among the non-oracle
 259 methods, especially when relocations are required ($L > 1$). MC+FCL’s low precision over-predicts
 260 obstructions: the planner fails for some scenes (lowest planning success), and the plans found have
 261 the largest mean and variance in length—indicating unnecessarily long, inefficient plans that depress
 262 execution at $L > 1$. Conversely, GRN’s plans rarely exceed $L = 1$, yet it attains the lowest execu-
 263 tion success with $L > 1$ cases failing almost entirely, showing its low recall causes frequent missed
 264 collisions and thus inaccurate plans.

265 6 Conclusion

266 We presented VORM, a model for object-wise grasp obstruction reasoning on point clouds in clut-
 267 tered environments. Central to VORM is representing not only the observed scene but also the ma-
 268 nipulator volume, its arm and gripper, as point clouds in a shared space, and learning their geomet-
 269 ric relationship via cross-attention; this lets VORM reason directly about where the robot collides
 270 with each object. This design yields reliable and balanced obstruction reasoning: whereas baselines
 271 sacrifice either precision or recall, VORM keeps both high, and grasp feasibility follows directly by
 272 aggregating its per-object predictions. The accuracy carries downstream—because VORM identifies
 273 the obstructing objects precisely, the planner removes exactly those that block the target, leading to
 274 higher target-retrieval success and shorter, more efficient relocation plans. Overall, VORM operates
 275 from partial observations, reasons over the full robot volume at arbitrary SE(3) grasps, and general-
 276 izes across clutter levels. By representing the robot as an occupancy point cloud, we expect VORM
 277 to apply to other embodiments. Extending it from single grasp poses to full motion trajectories is a
 278 promising direction for future work.

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